2:

Solution: No, consider $f : \mathbb{R} \to \mathbb{R}$ by $x \mapsto 0$. Take $A = \mathbb{R}$ and $x = 0$ so that x is a limit point of A but $f(A) = \{0\}$ which has no limit points since it is a singleton set and every neighborhood around 0 only contains 0. Thus $f(x) = 0$ is not a limit point of $f(A)$.

4:

Solution:

Proof. Without loss of generality we will show that $f : X \to X \times Y$ for $x \mapsto (x, y_0)$ is an embedding. It is clear that f is continuous since the sets $U \times V$ for $U \subseteq X$ open and $V \subseteq Y$ open form a basis for the product topology and

$$f^{-1}(U \times V) = \{x \mid (x, y_0) \in U \times V\} = \{x \in U \mid y \in V\}$$

which is U if $y_0 \in V$ and empty otherwise. Both sets are open in X and hence f is continuous. It is also obvious that f is injective as if $f(x_1) = f(x_2)$ then

$$(x_1, y_0) = (x_2, y_0) \Rightarrow x_1 = x_2.$$

Now to show that f^{-1} is continuous we first notice that $f(X) = X \times y_0$, and the projection $\pi_X(f(x)) = x$. So, the inverse $f(X) \to X$ is the restriction of π_X to the subspace $f(X)$, which is continuous.
An almost identical argument exists for $g : Y \to X \times Y$. □

8:

Solution:

(a) *Proof.* We first define

$$h : X \to Y \times Y, \quad x \mapsto (f(x), g(x))$$

and

$$C = \{(a, b) \in Y \times Y \mid a \leq b\} \subseteq Y \times Y$$

so that

$$h^{-1}(C) = \{x \mid f(x) \leq g(x)\}$$

We now show that C is closed in Y. Consider $Y \setminus C = \{(a, b) \mid a > b\}$. If there exists c such that $b < c < a$, then we take $U = (c, \infty)$ and $V = (-\infty, c)$ so for any $a' \in U$ and $b' \in V$ is it obvious that $(a', b') \in U \times V$. If there does not exist such c, we have $U = (b, \infty)$ and $V = (-\infty, a)$ so $a \in U$ and $b \in V$. Thus for $a' \in U$ and $b' \in V$ we have $a' > b$ and $b' < a$ but $a' \geq a$ and $b' \leq b$ and so

$$a' \geq a > b \geq b'$$

thus $(a', b') \in U \times V$. So, since U, V open and $U \times V$ is open it follows that $Y \setminus C$ is also open and thus C is closed in Y. Since f and g are continuous, h is also continuous and thus $h^{-1}(C)$ is closed in X. □

(b) *Proof.* Using (a) define $A, B \subseteq X$ closed by

$$A = \{x \mid f(x) \leq g(x)\} \quad \text{and} \quad B = \{x \mid f(x) \geq g(x)\}$$

It is clear that $X = A \cup B$ since by definition of orders every x either has $f(x) > g(x), g(x) < f(x)$, or $f(x) = g(x)$. Then, we note that on A, $\min\{f(x), g(x)\}$ and similarly on $B, \min\{f(x), g(x)\} = g(x)$

so that $h|_A = f(x)$ and $h|_B = g(x)$. Note that if for some x we have $f(x) = g(x)$ it doesn't matter whether h is restricted to A or B, since $x \in A \in B$ then $f(x) = g(x)$. Thus, by the pasting lemma, h is continuous. □

9:

Solution:

(a) *Proof.* Let $\{A_\alpha\}$ be finite and each A_α closed. Then,

$$X = \bigcup_\alpha A_\alpha$$

and $f|_{A_\alpha}$ is continuous and for $i \neq j f|_{A_{\alpha_i}} = f|_{A_{\alpha_j}}$ for all $x \in A_{\alpha_i} \cap A_{\alpha_j}$. Thus, by the pasting lemma we have that f is continuous by letting $f = f|_{A_{\alpha_i}}$ if $x \in A_{\alpha_i}$. □

(b) Let $X = [0,1]$ and $A_\alpha = \left[\frac{1}{\alpha}, 1\right] \cup \{0\}$. Clearly

$$X = \bigcup_\alpha A_\alpha$$

and each A_α is clsoed. Also $f|_{A_\alpha}$ is continuous since $f(x) = 1/x$ is clearly continuous on $\left[\frac{1}{\alpha}, 1\right]$, and 0 is an isolated point of A_α. As well, f is not continuous at 0 since for $x_n = 1/n, x_n \to 0$ but $f(x_n) \to \infty$ thus $f(x_n)$ does not converge to 0. Thus, each A_α is closed and there are only a countable A_α (seen by the map $n \mapsto A_n$ for $n \in \mathbb{N}$) but f is not continuous.

(c) *Proof.* Assume that $\{A_\alpha\}$ is locally finite and each A_α closed. It follows that $X \setminus \cup_\alpha A_\alpha$ is open since for $x \notin \bigcup A_\alpha$ there exists U open that intersects finitely many many sets $A_1, \ldots, A_k$. Then, take $V = U \setminus (A_1 \cup \cdots \cup A_k)$ open since the finite union of closed sets is closed and $x \in V$ by definition. Thus V misses every A_α since the infinite amount of A_α are already removed and the finite A_k are not included, thus ever point of the complement has an open neighborhood inside the complement. Pick som $C \subset Y$ and set

$$B_\alpha = (f|_{A_\alpha})^{-1}(C) = A_\alpha \cap f^{-1}(C)$$

Since $X \bigcup A_\alpha$ we know $f^{-1}(C) = \bigcup B_\alpha$ where B_α is closed in A_α because $f|_{A_\alpha}$ is continuous. It follows that B_α is closed in X since A_α is closed in X. Since $B_\alpha \subset A_\alpha$, any neighborhood that meets only finitely many A_α meets only finitely many B_α, thus making $\{B_\alpha\}$ locally finite. Thus $f^{-1}(C) = \bigcup B_\alpha$ is closed and f is continuous. □

2:

Solution:

Proof. Case 1: Assume both topologies have the box topology. Then, a basis for the box topology on X is

$$\{\Pi U_\alpha \mid U_\alpha \text{ open in } X\}$$

thus a basis for the subspace topology on Y is

$$\beta_1 = \{\Pi(A_\alpha \cup U_\alpha) \mid U_\alpha \text{ open in } X\}$$

A basis of the box topology on Y is

$$\{\Pi V_\alpha \mid V_\alpha \text{ open in } X\}$$

but each $V_\alpha = A_\alpha \cap U_\alpha$ for U_α open in X. Thus,

$$\beta_2 = \{\Pi(A_\alpha \cap U_\alpha) \mid U_\alpha \text{ open in } X\}$$

and $\beta_1 = \beta_2$ so their topologies are the same

Case 2: Assume both topologiers are the product topologies which have basis

$$\{\Pi U_\alpha \mid U_\alpha \text{ open in } X_a \text{ and } U_\alpha = X_\alpha \text{ for all but finitely many } \alpha\}$$

Now, consider a basis element of the subspace topology $Y \cap \Pi U_\alpha = \Pi(A_\alpha \cap U_\alpha)$. But, each $A_\alpha \cap U_\alpha$ is open in A_α, and for the α such that $U_\alpha = X_\alpha, A_\alpha \cap U_\alpha = A_\alpha \cap X_\alpha = A_\alpha$. Thus, this is a basis element of the product topology on Y. Take a basis element of the porduct topology on Y, ΠV_α where V_α is open in A_α and $V_\alpha \neq A_\alpha$ for finitely many α. For each $\alpha = 1, \dots, k$ such that $V_\alpha \neq A_\alpha$ choose U_α open in X_α with $V_\alpha = A_\alpha \cap U_\alpha$ and $U_\alpha = X_\alpha$ for all other $_\alpha$. Then ΠU_α is a basis element of the proudct topology on X, since $U_\alpha = X_\alpha$ for all but finitely many α. Also,

$$Y \cap \Pi U_\alpha = \Pi(A_\alpha = U_\alpha) = \Pi V_\alpha$$

for $\alpha = 1, \dots, k$. This is true by choice of U_α and for all other α, $A_\alpha \cap X_\alpha = A_\alpha = V_\alpha$. Thus, ΠV_α is open in the subspace topology and each topology contains each other's basis elements, and thus are equal. □

6:

Solution:

Proof. ($\Rightarrow$) Let U be a neighborhood of $\pi_\alpha(x)$ in X_α. Then $\pi_\alpha^{-1}(U)$ is open in the product by being a subbasis element, and it contains x. Since $x_i \to x$, we have that $x_i \in \pi_\alpha^{-1}(U)$ for i large, so $\pi_\alpha(x_i) \in U$ also for i large.
($\Leftarrow$) Let ΠU_α be a basis element containing x with $U_\alpha = X_\alpha$ for all but finite α. Then for each j, $\pi_{\alpha_j}(x_i) \to \pi_{\alpha_j}(x)$ and U_{α_j} is a neighborhood of π_{α_j}, so there is N_j with $\pi_{\alpha_j}(x_i) \in U_{\alpha_i}$ for $i \geq N_j$. We take $N = \max\{N_1, \dots, N_k\}$. For $i \geq N$ every $x_i \in U_\alpha$ for some α so $x_i \in \Pi U_\alpha$ and $x_i \to x$. □

In the box topology on $\mathbb{R}^\omega$ take $x_i = \left(\frac{1}{i}, \frac{1}{i}, \dots\right)$ and $x = (0, 0, \dots)$. Then $\pi_n(x_i) = \frac{1}{i} \to 0$ but $U = \Pi\left(-\frac{1}{n}, \frac{1}{n}\right)$ is a box neighborhood of 0 and $x_i \in U$ implies $\frac{1}{i} < \frac{1}{n}$ for all n which is impossible.

7:

Solution: The closure of $\mathbb{R}^\infty$ is $\mathbb{R}^\omega$ in the product topoloy and $\mathbb{R}^\infty$ in the box topology.

Proof. Let $x = (x_1, x_2, \dots) \in R^\omega$ and let ΠU_n be a basis element containg x, with $U_n = \mathbb{R}$ for $n > N$. Take $y = (x_1, \dots, x_n, 0, 0, \dots)$ so $y \in \mathbb{R}^\infty$, and $y \in U_n$ for all n (for $n \leq N, y_n = x_n$ and for $n > N$ since $U_n = \mathbb{R}$). So, every neighborhood of x meets $\mathbb{R}^\infty$, and the closure of $\mathbb{R}^\infty$ is all of $\mathbb{R}^\omega$.

We now show $\mathbb{R}^\infty$ is closed by showing its complement is open. Consider $x \notin \mathbb{R}^\infty$ so $x_n \neq 0$ for infinitely many n. Define

$$U_n = \begin{cases} (x_n/2, 3x_n/2) & \text{if } x_n \neq 0 \\ \mathbb{R} & \text{if } x_n = 0 \end{cases}$$

Then ΠU_n is a box-neighborhood of x Any y in it has $y_n \neq 0$ whenever $x_n \neq 0$ which is infinitely many n so $y \neq \mathbb{R}^\infty$. Hence the complement is open so $\mathbb{R}^\infty$ is closed and thus it is its own closure. □